neutrons, too, are sure to have been liberated. Figure 25B shows a similar, still more complicated process. In the disintegration of one nucleus a particle has been liberated, which subsequently induces a transmutation reaction in a second nucleus. The difference in the density of the silver grains in the individual tracks is due to the difference in the velocities of the nuclear particles. The faster they are, the more thinly they are spread, and the longer is the track which they leave. Of course, actually, the particles are dispersed in all directions, and therefore, the majority of their paths appear more or less foreshortened.

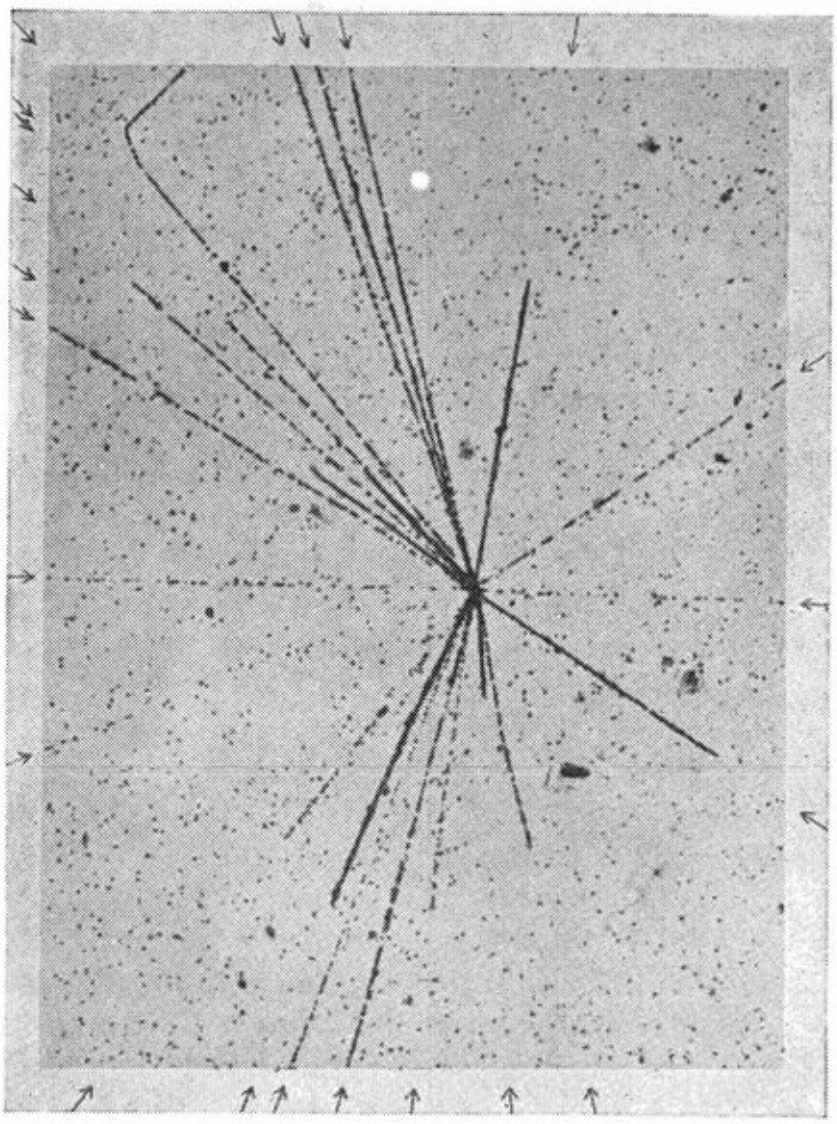


Figure 25A.—Disintegration of a nucleus by cosmic ray particle of very big energy (*after Powell and Occhialini*).